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Olumide Erinfolami

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Machine Learning Engineer

Machine Learning Engineer focused on fintech and tabular ML: credit risk, payment recovery and forecasting. I build end-to-end systems that pair well-calibrated models with cost-based decisions, explain every prediction, and ship as tested software behind a REST API. Comfortable from survival and scorecard models through gradient boosting, time series and deep learning.

TECHNICAL SKILLS

Languages Proficient: Python, SQL • Experience: JavaScript, HTML/CSS

Tech Stack scikit-learn, LightGBM, XGBoost, CatBoost, Optuna, SHAP, PyTorch (GRU/LSTM), statsmodels, SciPy, pandas, NumPy, Flask, FastAPI, React, pytest, GitHub Actions (CI), Streamlit, Git/GitHub

EXPERIENCE

Machine Learning Intern, Notzero Technologies — Mar 2026 – Sep 2026

Recoup: payment-recovery ML system (GitHub) — *Nigeria (Hybrid)*

- Built **Recoup**, a failed-payment recovery system that decides when to retry a failed subscription payment, how many times, and when to ask for a new card, replacing the fixed "retry after 1, 3 and 7 days" rule.
- Modeled each customer's chance of paying at each retry time with a hierarchical mixture-cure hazard model in Python and SciPy, sharing strength across merchants so small businesses with little history still get good estimates.
- In simulation, recovered **31% more revenue per failed payment** than the fixed rule ($15.93 vs. $12.12), with gains of 20–35% across six stress-test scenarios and +19% for small merchants.
- Wrote an exact planner that values every candidate retry schedule in closed form, with Thompson-sampling exploration logged for doubly-robust off-policy evaluation and a safety gate that keeps the old schedule in charge until evidence confirms the gain.
- Shipped it as working software: an ops CLI and a FastAPI + React operator console.

PROJECTS

Can I borrow.ai: Loan Default Risk Engine | *LightGBM, scikit-learn, SHAP, Flask, pytest* — GitHub

- Built an end-to-end system that predicts whether a borrower will default, trained on 255K consumer loans; the pipeline fits and calibrates logistic regression and LightGBM and writes versioned, auditable models with auto-generated model cards.
- Reached **ROC-AUC 0.76** on 51K unseen loans with well-calibrated risk estimates (ECE 0.004; predicted default rate 11.70% vs. actual 11.61%); a cost-tuned approval threshold **catches 80% of defaults**.
- Explains every decision with its top SHAP reasons, keeps protected attributes out of the model, checks for unfair outcomes on every run, and serves it all through a Flask API and web app with 93 tests.

CrediPulse: Explainable Credit Scoring Platform | *WOE scorecard, LightGBM, Optuna, SHAP, Flask, JavaScript* — GitHub

- Built an auditable credit scorecard: sign-constrained WOE logistic regression whose per-feature points add up exactly to the score, so they double as reason codes for declines.
- Set approval cutoffs in money, P(default) × EAD × LGD per loan-size band, and reported metrics with bootstrap confidence intervals and a DeLong test against a tuned LightGBM challenger (**ROC-AUC 0.81**).

Financial Time-Series Forecasting: Oil, Inflation & FX | *statsmodels, arch, LightGBM, PyTorch* — GitHub

- Forecast 39 years of Brent crude volatility: gradient boosting on lag features reached **4.6% error vs. 60%** for the classical GARCH(1,1) model.
- Benchmarked ARIMA, SARIMA, LightGBM and LSTM on US CPI inflation with walk-forward validation, and built GRU/LSTM models in PyTorch for EUR/USD volatility against the HAR-RV baseline.

EDUCATION

Covenant University, B.Sc. Computer Science — 2021 – 2025

EXTRACURRICULAR

- **DSN Bootcamp Qualification Hackathon 2026, placed 27th:** under repeated paired 5-fold cross-validation, a simple explainable store-level model (RMSE 1,070.8) beat tuned LightGBM, XGBoost and CatBoost; shipped as a tested Python package with GitHub Actions CI.